



Introduction

Data Science and Management

Corso di Laurea Magistrale in Ingegneria Gestionale

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- Course Program
- Course Organization
- Exam
- Software
- Books
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Course Programme

- Python recap
 - Pandas, numpy, matplotlib, scikit-learn, keras,...
- Statistics background
- Data exploration and visualization
- Machine learning and data mining

Course Organization

Class lectures and code sessions

- **All the teaching material available on Github**

Schedule

- Tuesday, 14:00-18:00
- Thursday, 14:00-17:00 / 18:00

Additional notes

- Total amount of hours: **81**
- The course should finish **by mid December**

Course Organization



Office hours

- Online, using Teams or Google Meet
- Please send me an e-mail in advance
- You can also ask me (quick) questions via e-mail

Please **subscribe** to the Moodle website!

- I will send communications through the platform
- Changes in lecturing timeline, new slides, etc...

Exam



The exam will consist of **two parts**:

- Project (weight ~50%)
- Oral exam (weight ~50%)

Exam



About the project (more info throughout the course):

- You can work in teams of **up to 3 people**
- We will assign a project to each student / team, or you can propose a topic
- The project requires:
 - **Code:** implementation or experimentation
 - **Report:** 4-pages document in english presenting the problem, the solutions and the results. It is important to **motivate** the choices you made
 - **Presentation:** slides for 10 mins project presentation

Oral exam will start with project discussion followed by questions about course's topics

- **Questions** already available!

Dates

- Dates to be confirmed:
- January, February, April, June, September, November

I will use it during the course

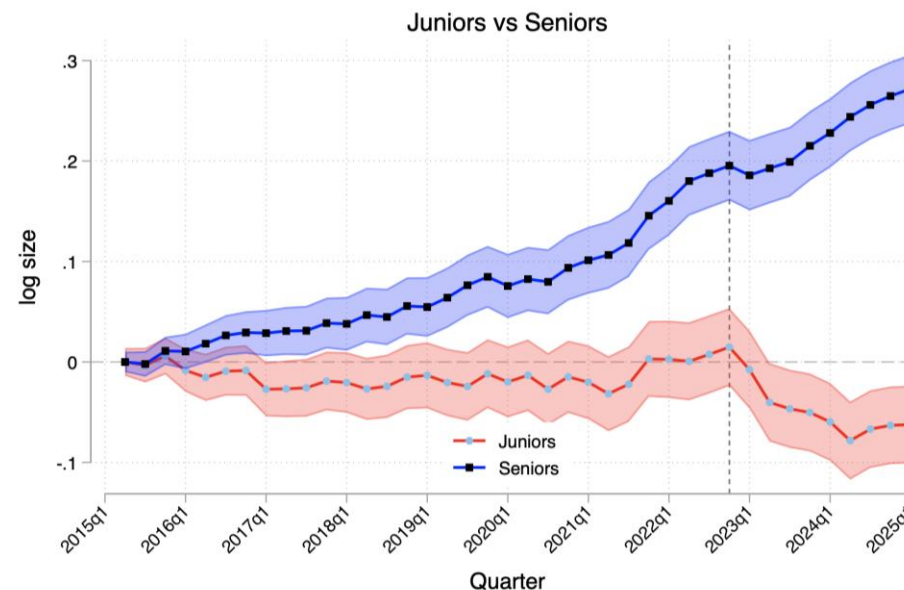
You can use it for the exam, **however** you must understand (and be able to explain) the code it produces and the techniques (models) being used.

Not only code generation:

- Explanation / examples
- Code comments and refactoring

AI is a capability multiplier, not a substitute. Those who use it well become more valuable at work; those who use it poorly risk reducing their own relevance.

Focus on skills and quality.



Software

For the data science part, we will use Python 3+

Available for Windows, Linux, MacOSX

You can use any IDE you like: during the course - I will use VS Code.

<https://www.python.org/downloads/>

<https://code.visualstudio.com/download>

<https://github.com/settings/education/benefits> (copilot)

Books



- J. VanderPlas. Python Data Science Handbook, O'Reily
- A. Géron. Hands-On Machine Learning with Scikit-Learn, Keras, and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reily

Schedule

1	Intro
2	Python recap and count
3	Python recap and count + plot
4	Python + Database
5	ERP + sistemi informativi (MES...)
6	Big Data
7	Statistics
8	Statistics
9	Statistics
10	Item set
11	Clustering
12	ML Intro + KNN
13	Trees
14	Trees + Timeseries
15	Linear Models
16	NN
17	Conv NN
18	AutoEncoder
19	RNN
20	LLM + Transformer
21	Digital Twin
22	Federated Learning